

Lesson 4-4 . Period 1

Adding & Subtracting Rational Expressions +
Resistance DOK-3 (Q#15 Prep)

Klimsara-adapted . Student-centered . No Blooket

Do Now — Critique & Explain

Algebra 2 · Lesson 4-4 ·

Single-Period Quick

DOK 1

5 min

bridge from 4-3

Do Now (4-4-savvas-model-discuss-lesson-4-4-launch) · Common denominator conflict

Teo claims that a common denominator of $\frac{1}{2} + \frac{1}{3} + \frac{1}{9}$ is $2 + 3 + 9 = 14$. Evander claims it is $2 \cdot 3 \cdot 9 = 54$.

- A.** Is either student correct? Explain using the word *common denominator*.
- B.** Find the sum. Explain your method.
- C.** Timothy says: multiply denominators together. Always

Launch — Rules + Example 3 + Example 4 (teacher models)

Algebra 2 · Lesson 4-4 · Single-Period Quick

DOK 1–2

13 min

teacher models

Rules — Add/Subtract Rational Expressions (keep printed)

1. LIKE denominators: combine numerators, keep denominator, simplify.
2. UNLIKE denominators: (a) factor each denominator, (b) find the LCM, (c) rewrite each fraction with LCM as denominator, (d) combine numerators, (e) simplify.
3. LCM = every DISTINCT factor at its highest power across both denominators.
4. Always check: can the final expression simplify by cancelling a

Explore — TPS + DOK-3 Capstone

Algebra 2 .

Lesson 4-4 · Single-Period Quick

DOK 2–3

32 min

student work

6 items in order. Plan ~10 min for Practice #32 (Resistance). Wed 45-min variant: skip #16 + #26.

Try It 3 · Add unlike denominators Find the sum of two rational expressions with unlike denominators. Factor first; build the LCM.

Try It 4 · Subtract unlike denominators Simplify. Watch: minus sign distributes across EVERY term of the second numerator.

DOK 2

5 min

DOK-3 reveal + Q#15 bridge

Sentence frame · Resistance (#32) “For the parallel circuit, $1/R_{\text{total}} = 1/R_1 + 1/R_2$. I found a common denominator of _____, added the reciprocals to get $1/R_{\text{total}} = \text{_____}$, then flipped to get $R_{\text{total}} = \text{_____}$. The ratio R_A/R_B simplified to _____ because _____.”

Bridge to Topic 4 LEHS Q#15 The resistance move

Exit Ticket — Summary Recap

Algebra 2 · Lesson 4-4 ·

Single-Period Quick

DOK 1–2

3 min

not graded

Complete on your exit ticket

1. To add rational expressions with unlike denominators, I must first find the _____ of the denominators.
2. Practice #32 (resistance) showed me that reciprocal-sum problems (like $1/R_1 + 1/R_2$) use the same algebraic steps as adding rational expressions because _____.